

EXPERIMENT 12: TIC TAC TOE GAME

Aim

To write a Python program to implement a **Tic Tac Toe game**.

Objective

- To understand game-state representation.
- To implement a two-player game.
- To check winning conditions.
- To handle player turns.
- To identify win, loss, and draw conditions.

Algorithm

1. Initialize a 3×3 board with empty spaces.
2. Assign X to Player 1 and O to Player 2.
3. Display the board.
4. Ask the current player to enter a position.
5. Check whether the selected position is empty.
6. Place the player's symbol.
7. Check all rows, columns, and diagonals for a winner.
8. If a player wins, display the winner and stop.
9. If all positions are filled, declare a draw.
10. Otherwise, switch the player and continue.

Flowchart

START

|

v

Initialize Board

|

v

Display Board

|

v

Get Player Move

|

v

Position Empty?

/ \

NO YES

| |

| v

| Place Symbol

| |

| v

| Winner?

| / \

| YES NO

| | |

| v v

| Display Board Full?

| Winner / \

| YES NO

| | |

| v v

| Draw Change Player

| |

+-----+

|

v

END

Python Program

Writing

Tic Tac Toe Game

```
board = [" "] * 9
```

```
def display():
```

```
    print()
    print(board[0], "|", board[1], "|", board[2])
    print("--+---+--")
    print(board[3], "|", board[4], "|", board[5])
    print("--+---+--")
    print(board[6], "|", board[7], "|", board[8])
    print()
```

```
def check_winner(player):
```

```
    win = [
        (0, 1, 2),
        (3, 4, 5),
        (6, 7, 8),
        (0, 3, 6),
        (1, 4, 7),
        (2, 5, 8),
        (0, 4, 8),
        (2, 4, 6)
```

```
    ]
```

```
    for a, b, c in win:
```

```
        if board[a] == board[b] == board[c] == player:
```

```
            return True
```

```
    return False
```

```
player = "X"
```

```
for i in range(9):
```

```
    display()
```

```
    pos = int(input("Enter position (1-9): ")) - 1
```

```
    if board[pos] != " ":
```

```
        print("Position already occupied!")
```

```
        continue
```

```
    board[pos] = player
```

```
    if check_winner(player):
```

```
        display()
```

```
        print(player, "wins!")
```

```
        break
```

```
    if i == 8:
```

```
        display()
```

```
        print("Draw!")
```

```
        break
```

```
    if player == "X":
```

```
        player = "O"
```

```
    else:
```

```
        player = "X"
```

Sample Output

```
| |  
--+---+--  
  
| |  
--+---+--  
  
| |
```

Enter position (1-9): 1

```
X| |  
--+---+--  
  
| |  
--+---+--  
  
| |
```

Enter position (1-9): 5

```
X| |  
--+---+--  
  
| O |  
--+---+--  
  
| |
```

Enter position (1-9): 2

```
X| X|  
--+---+--  
  
| O |  
--+---+--  
  
| |
```

Enter position (1-9): 8

```
X | X |  
--+---+--  
| O |  
--+---+--  
| O |
```

Enter position (1-9): 3

```
X | X | X  
--+---+--  
| O |  
--+---+--  
| O |
```

X wins!

Result

Thus, the **Tic Tac Toe** game was successfully implemented in **Python** with player turns and winning conditions.